

Day 69 – Access Control Lists (ACLs)

Overview

Day 69 introduced **Access Control Lists (ACLs)** and marked the transition from pure networking into network security.

In previous labs, the focus was:

- Day 65 → RIP (Routing)
- Day 66 → DHCP (IP Assignment)
- Day 67 → DNS (Name Resolution Concepts)
- Day 68 → NAT (Private-to-Public Connectivity)

Day 69 introduced a new question:

Just because devices can communicate, should they be allowed to?

ACLs provide the ability to permit or deny traffic based on defined security rules.

Objectives

By the end of this lab, I was able to:

- Understand the purpose of ACLs.
 - Configure a standard ACL.
 - Block traffic from a specific source network.
 - Apply ACLs to router interfaces.
 - Understand the difference between permitted and denied traffic.
 - Verify ACL functionality using connectivity tests.
 - Understand the relationship between ACLs, AWS Security Groups, and Network ACLs.
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Network Topology

The Day 68 topology was maintained.

LAN10 LAN20 LAN30

PC0 PC1 PC2

|||

SW0 SW1 SW2

|||

R0 ----- R1 ----- R2 ----- Router3

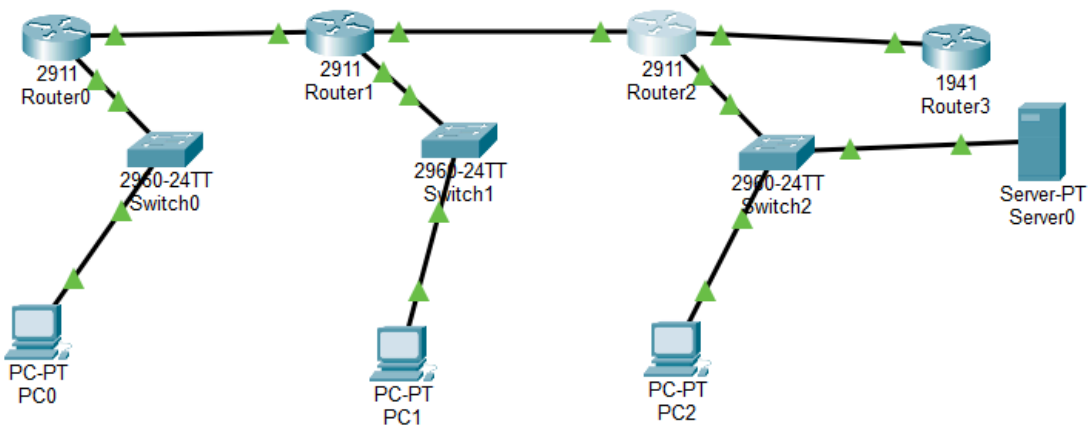
|

Server0

Services already configured:

- ✓ RIP
- ✓ DHCP
- ✓ NAT
- ✓ Server0
- ✓ ISP Simulation

Screenshot #1 – Full Day 69 Topology



Highlight

- Four routers
 - Three switches
 - Three PCs
 - Server0
 - ISP Router
 - NAT Configuration
-

ACL Fundamentals

What is an ACL?

ACL stands for:

Access Control List

An ACL acts like a security guard for a router.

When traffic arrives:

Packet

↓

ACL Inspection

↓

Permit or Deny

The router determines whether traffic is allowed to continue.

Lab Goal

The objective was:

Block LAN10

from reaching

LAN30

while still allowing:

LAN20

to reach

LAN30

This demonstrates traffic filtering based on source networks.

Pre-ACL Behavior

Before ACL implementation:

PC0 → PC2



PC1 → PC2



All LANs could communicate freely.

ACL Configuration

A standard ACL was created on Router2.

```
access-list 10 deny 192.168.10.0 0.0.0.63
```

```
access-list 10 permit any
```

Explanation

Deny Statement

```
access-list 10 deny 192.168.10.0 0.0.0.63
```

Blocks:

192.168.10.0/26

which corresponds to:

LAN10

Examples:

192.168.10.2

192.168.10.10

192.168.10.50

Permit Statement

```
access-list 10 permit any
```

Allows all remaining traffic.

Without this command, the ACL would eventually deny all traffic because every ACL contains an implicit:

deny everything

at the end.

Screenshot #2 – ACL Creatio

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#access-list 10 deny 192.168.10.0 0.0.0.63
Router(config)#
Router(config)#access-list 10 permit any
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

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access-list 10 deny 192.168.10.0 0.0.0.63

access-list 10 permit any

ACL Placement

The ACL was applied on:

Router2

specifically:

interface g0/0

ip access-group 10 out

Why Router2?

LAN30 sits behind:

Router2 G0/0

Every packet attempting to enter LAN30 must leave through:

G0/0

Applying the ACL there protects LAN30.

Why OUTBOUND?

Traffic path:

PC0

↓

R0

↓

R1

↓

R2

↓

G0/0

↓

PC2

The packet exits Router2 through:

G0/0

Therefore:

`ip access-group 10 out`

filters packets before they enter LAN30.

Screenshot #3 – ACL Applied to Interface

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)#
Router(config-if)#ip access-group 10 out
Router(config-if)#
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Highlight

interface g0/0

ip access-group 10 out

Security Logic

After applying the ACL:

LAN10

PC0

↓

PC2

 Denied

LAN20

PC1

↓

PC2

 Allowed

Verification

Test 1

From PC0:

ping 192.168.30.2

Expected Result:

Failed

This proved:

LAN10

was blocked.

Test 2

From PC1:

ping 192.168.30.2

Expected Result:

Success

..

This proved:

LAN20

remained permitted.

Screenshot #4 – PC0 Blocked

```
C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 10.0.0.6: Destination host unreachable.
Reply from 10.0.0.6: Destination host unreachable.
Reply from 10.0.0.6: Destination host unreachable.
Reply from 10.0.0.6: Destination host unreachable.

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

 INSERT SCREENSHOT HERE

Highlight

Request Timed Out

Screenshot #5 – PC1 Successful Ping

```
Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=126
Reply from 192.168.30.2: bytes=32 time=7ms TTL=126
Reply from 192.168.30.2: bytes=32 time<1ms TTL=126
Reply from 192.168.30.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 2ms

C:\>
```

Highlight

Reply From
192.168.30.2

ACL Verification

Router2:

show access-lists

Output should show:

Standard IP access list 10

deny 192.168.10.0

permit any

,

Hit counters increase as packets are processed.

Screenshot #6 – ACL Counters

```
Router#show access-lists
Standard IP access list 1
  10 permit 192.168.0.0 0.0.255.255
Standard IP access list 10
  10 deny 192.168.10.0 0.0.0.63 (4 match(es))
  20 permit any (4 match(es))
Router#
```

Highlight

deny 192.168.10.0

Hit count increasing.

Why We Used Router2 Instead of Router0

Standard ACLs only inspect:

Source Addresses

Cisco best practice:

Place Standard ACLs
near the destination.

Destination:

LAN30

Router closest to LAN30:

Router2

Therefore Router2 is the correct placement.

AWS Connection

ACLs directly relate to:

AWS Security Groups

Example:

Allow HTTP

Allow HTTPS

Deny Everything Else

AWS Network ACLs

Example:

Allow LAN20

Deny LAN10

The logic is nearly identical.

What I Learned

- ACLs control who can communicate.
 - Standard ACLs filter based on source addresses.
 - ACLs are processed top-to-bottom.
 - Every ACL has an implicit deny statement.
 - ACL placement affects network behavior.
 - Network security depends on traffic control.
 - ACLs are a foundation for cloud security.
 - AWS Security Groups and NACLs use similar concepts.
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Reflection

Day 69 introduced Access Control Lists and demonstrated how security policies are enforced within routed networks. Prior labs focused on enabling communication, while this lab focused on restricting communication. By configuring a standard ACL on Router2, I successfully blocked LAN10 from reaching LAN30 while still allowing LAN20 access. This lab reinforced the concept that network security is not about making communication possible, but about controlling which communication is permitted.
